Microsoft Elevate

CAPSTONE PROJECT

Smart Student Performance Insights Dashboard

PRESENTED BY

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Introduction :

- Education institutions collect large amounts of student performance data. However, analyzing this data manually is difficult and time consuming.
- A dashboard helps visualize student performance using charts and graphs, making it easier for teachers and administrators to understand patterns and trends.
- This project focuses on creating a **Student Performance Analysis Dashboard** to analyze student scores effectively.

Problem Statement :

- Educational institutions store large volumes of student data.
- Problems include:
- • Difficult to analyze student performance manually
  • Hard to identify performance patterns
  • Teachers cannot easily compare scores
  • Lack of visual representation of data
- A data visualization system is required to solve these problems.

Proposed Solution :

- The proposed solution is to develop a **Student Performance Analysis Dashboard** that helps educators analyze academic performance using data visualization techniques.
- The dashboard provides a clear and interactive interface where users can easily understand student performance through graphical representations.
- The system helps to:
- • Visualize student scores using charts and graphs
  • Compare performance across different categories such as gender, ethnicity, and parental education
  • Identify trends and patterns in academic performance
  • Analyze subject-wise performance including math, reading, and writing
  • Monitor average scores and overall performance levels
  • Provide quick insights using interactive dashboard elements
- The dashboard also includes filtering options that allow users to analyze specific data segments. This makes it easier for teachers and administrators to explore the data and identify areas where students need improvement.
- By using data visualization techniques, the system transforms raw data into meaningful insights that support better academic planning and decision making.

Dataset Explanation :

- The dataset contains student academic performance information.
- Key attributes include:
- • Student ID
  • Gender
  • Ethnicity
  • Parental Level of Education
  • Test Preparation Course
  • Math Score
  • Reading Score
  • Writing Score
- This dataset is used to analyze and visualize student performance patterns.

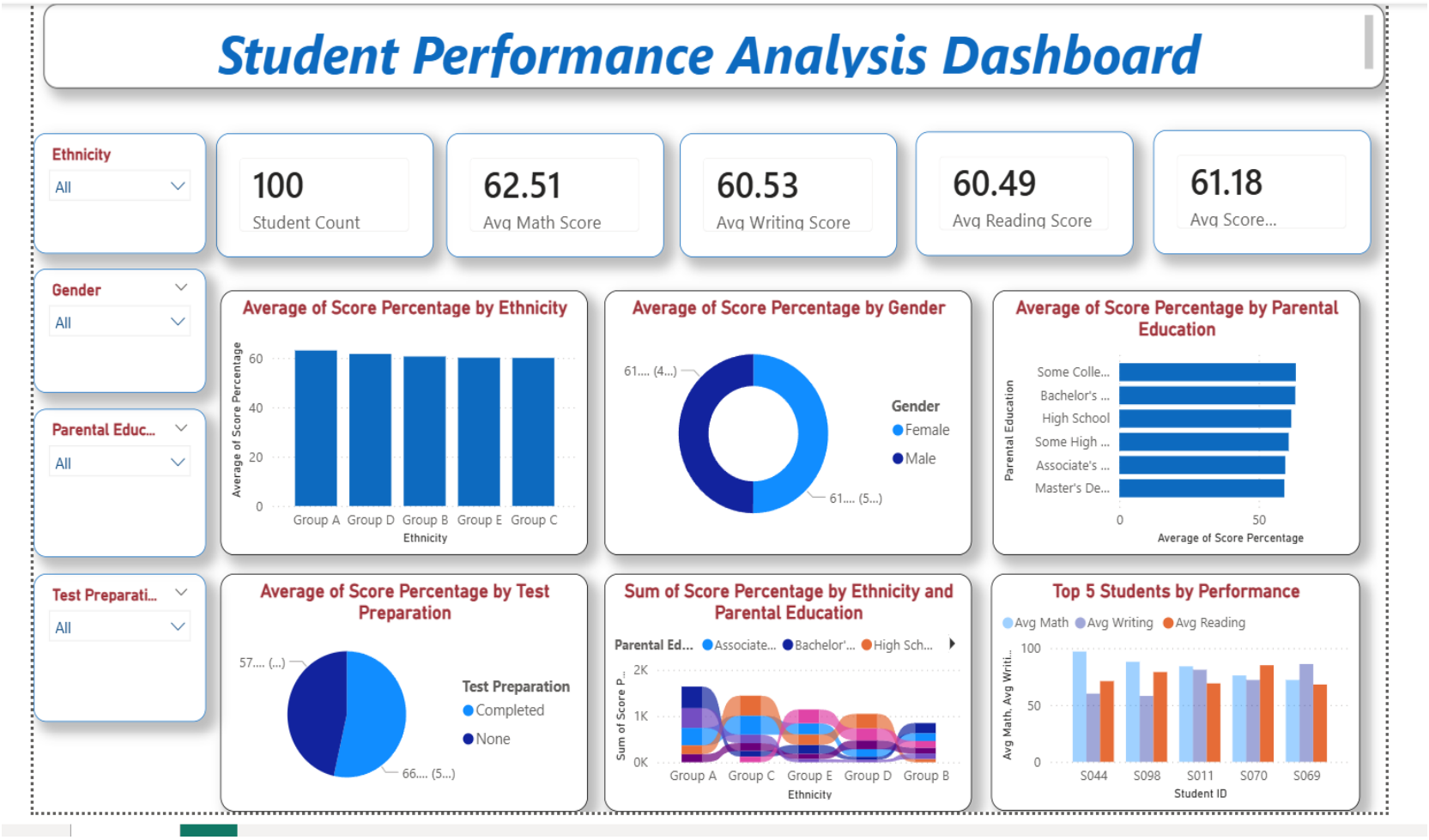
Technology Stack:

- The following technologies were used to develop the project:
- Data Source
  Student Performance Dataset (Excel)
- Data Processing
  Microsoft Excel
- Visualization Tool
  Power BI
- Charts Used
- • Bar Chart
  • Pie Chart
  • Donut Chart
  • KPI Cards
  • Performance Comparison Charts

System Development Approach :

- The project follows the following steps:
- Data Collection
  Student dataset collected from Excel.
- Data Cleaning
  Unnecessary values removed and data formatted.
- Data Analysis
  Scores analyzed based on gender, ethnicity and parental education.
- Dashboard Creation
  Interactive charts created using Power BI.
- Performance Visualization
  Dashboard shows student performance insights.

Dashboard :



The dashboard includes:

- Student count KPI
- Average scores
- Gender comparison
- Ethnicity comparison
- Test preparation impact

Results :

- The dashboard successfully visualizes student performance data.
- Key outcomes:
- • Easy understanding of student score patterns
  • Identification of high performing students
  • Comparison of performance between categories
  • Interactive filtering improves analysis
- This helps educators make better academic decisions.

Conclusion :

- The Student Performance Analysis Dashboard provides an effective way to analyze student data.
- The system helps educators identify trends and evaluate academic performance quickly.
- Using data visualization tools like Power BI improves decision making and enhances educational insights.

Future Scope :

- Future improvements may include:
- • Real-time student data integration
  • AI-based performance prediction
  • Web-based dashboard deployment
  • Integration with school management systems

References :

- Microsoft Power BI Documentation
- Student Performance Dataset
- Data Visualization Techniques

GITHUB LINK :

Thank You